

A NEGATIVE RESULT FOR TRAINING-FREE TEMPORAL VISUAL CACHING IN OPENVLA-OFT

Ved Dwivedi^{1,2} Cheng Yang² Bo Yuan²

¹John P. Stevens High School, Edison, New Jersey

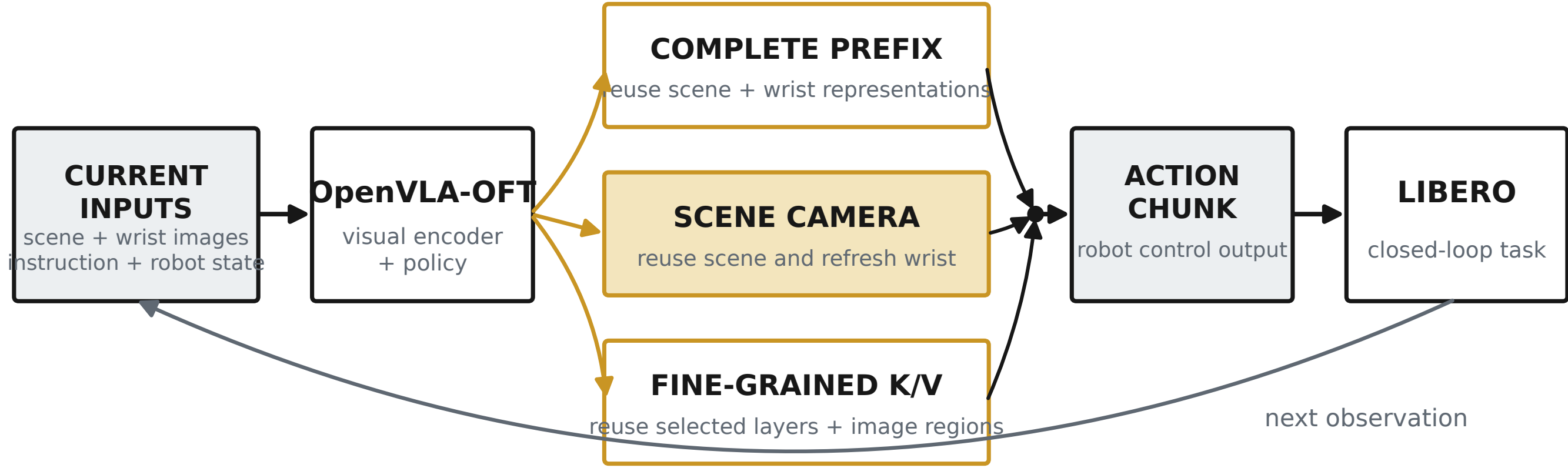
²Department of Electrical and Computer Engineering, Rutgers University-New Brunswick, New Jersey

INTRODUCTION

- Vision-language-action policies repeatedly process two camera views during robot control.
- Consecutive observations are often similar, but reused representations may omit small changes that affect the next action.
- We test whether three training-free reuse boundaries can preserve task success while reducing complete-query latency in closed-loop LIBERO tasks.

METHODS

HOW TEMPORAL VISUAL REUSE WAS EVALUATED



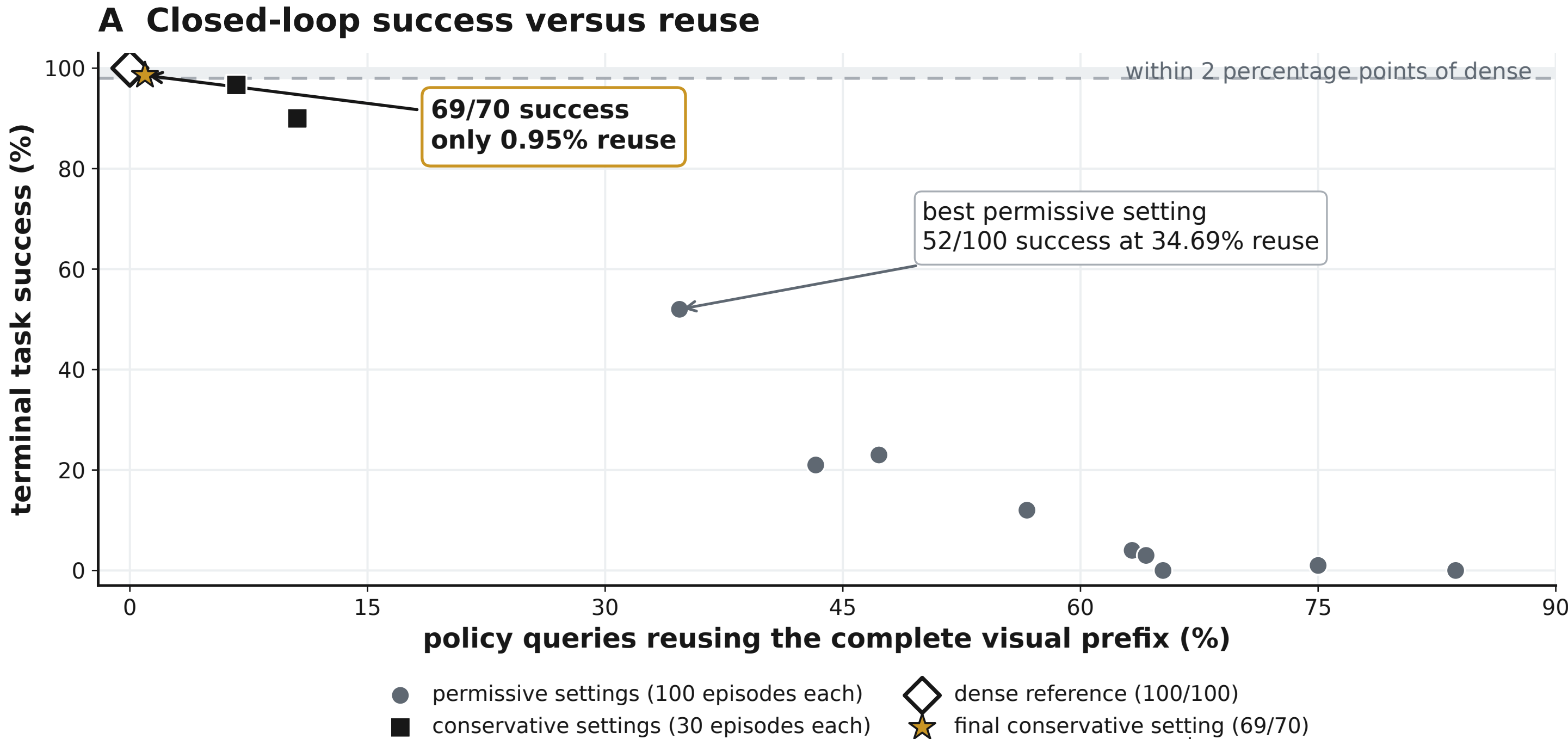
Actions determine the next observation, so every strategy was evaluated in closed loop.

OpenVLA-OFT 7B • LIBERO • NVIDIA TITAN RTX • task success • query wall time

RESULTS

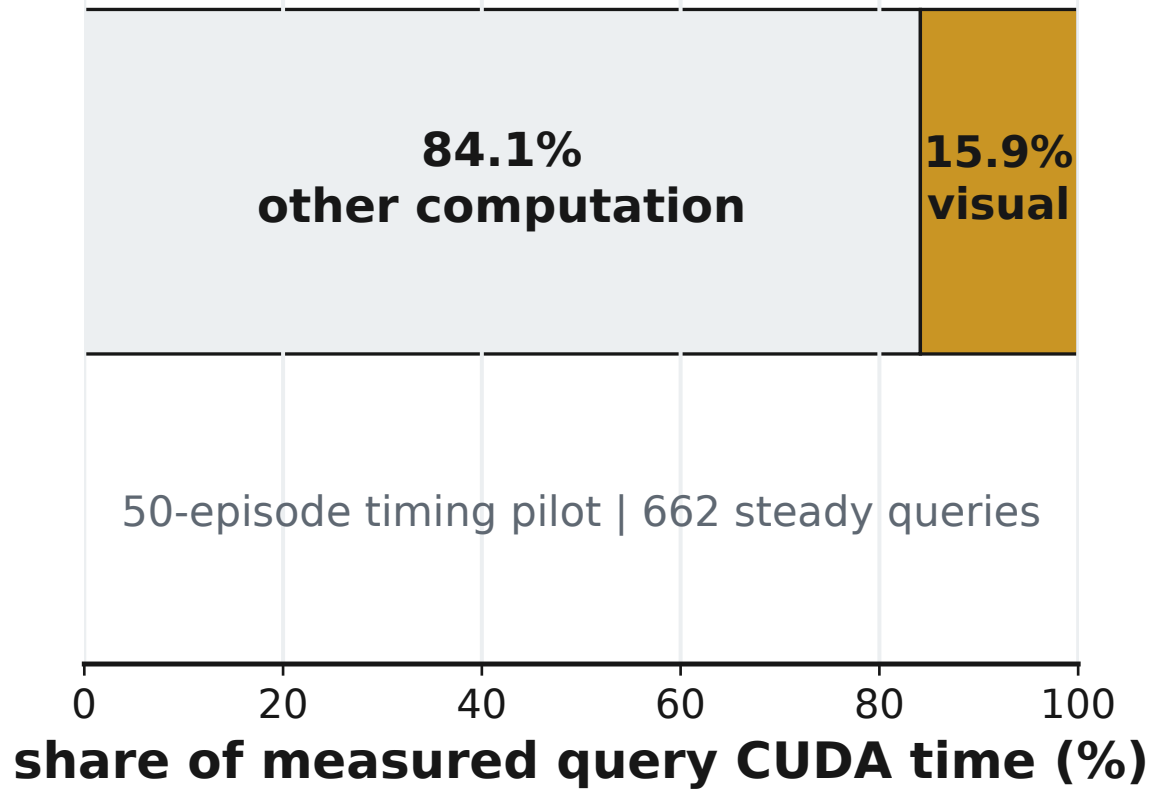
COMPLETE VISUAL-PREFIX REUSE

1,160 primary closed-loop episodes | separate 50-episode timing pilot



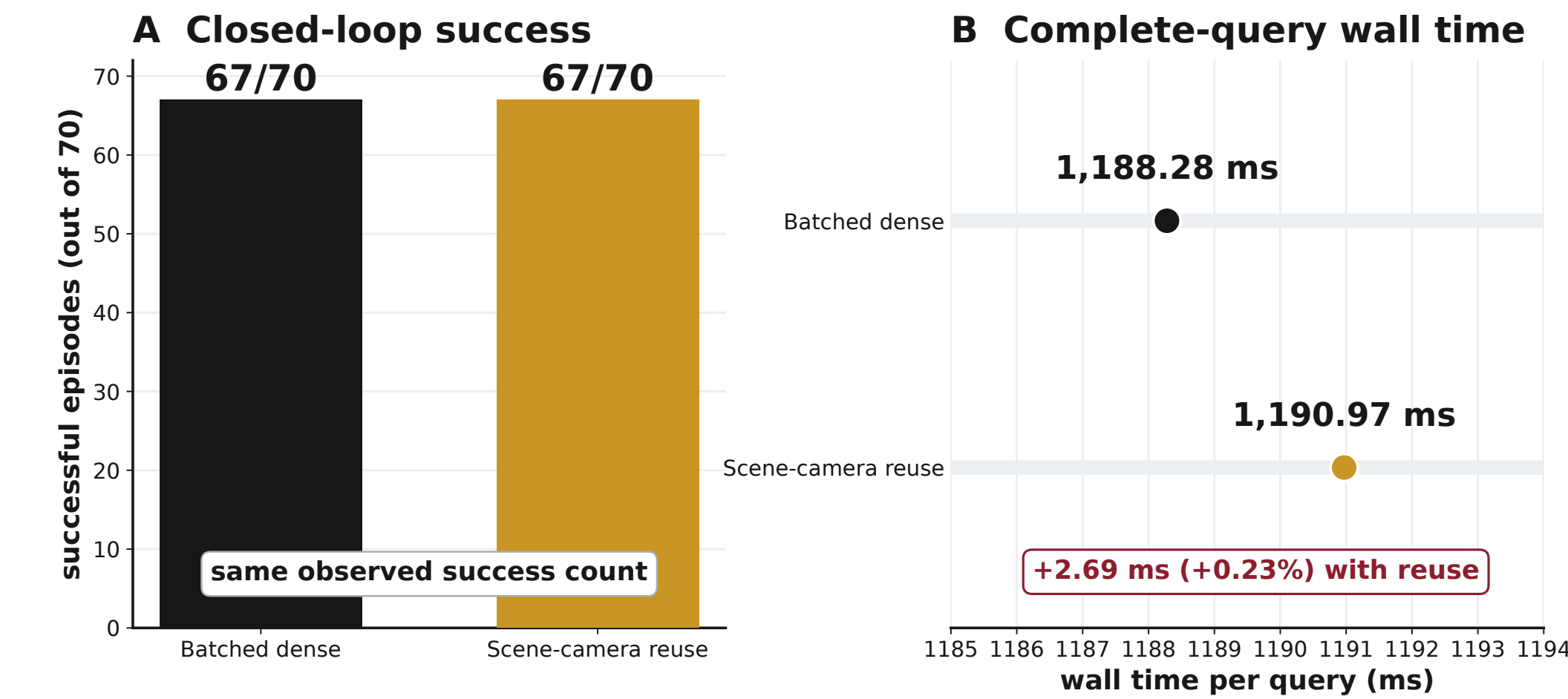
B Measured CUDA-time budget

Even zero-cost visual processing at most 1.189x theoretical speedup



SCENE-CAMERA REUSE

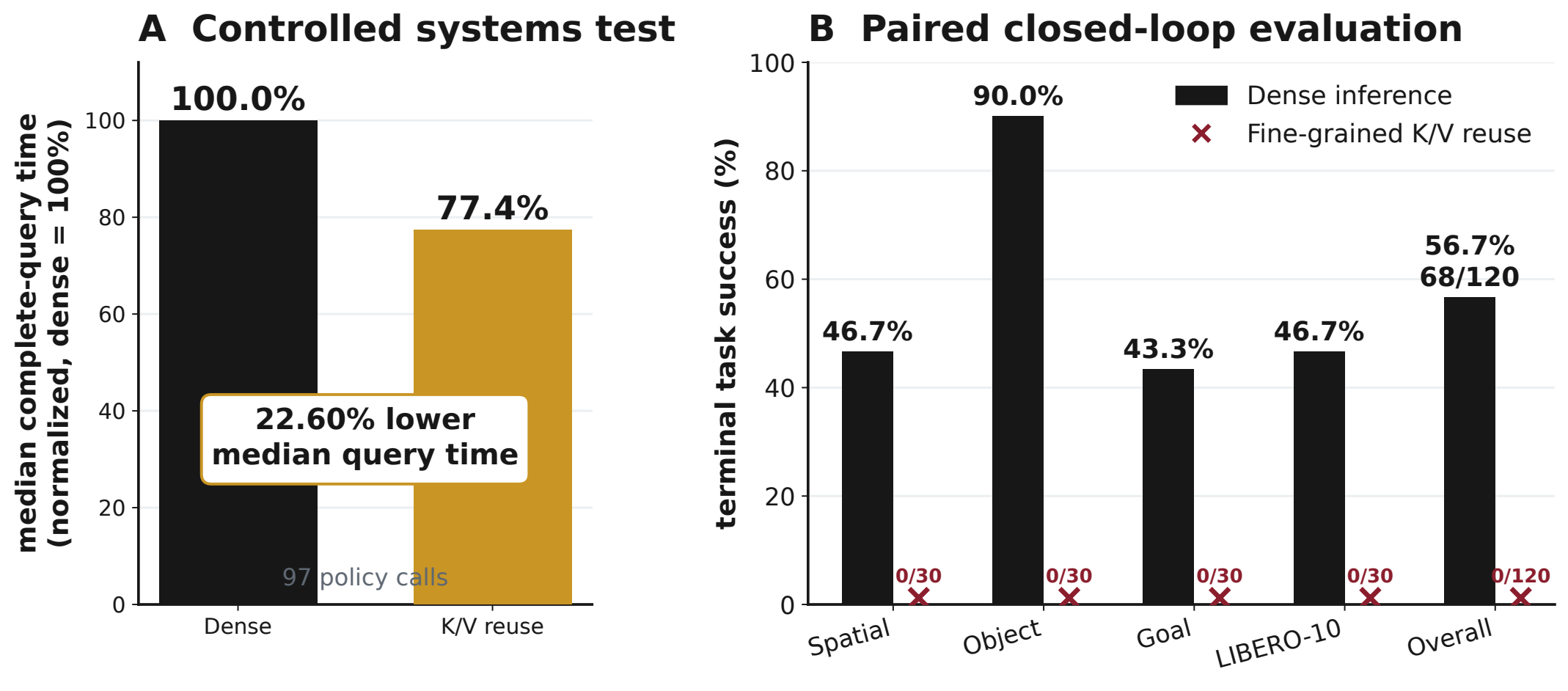
Same observed success count, but no end-to-end speedup



Scene representation reused on 25.24% of queries | 70 episodes per condition from paired initial states

FINE-GRAINED K/V REUSE

Lower query time did not preserve closed-loop task success



Closed-loop evaluation with 30 episodes per suite and condition (120 overall)

Timing and task success were measured in separate, explicitly labeled tests.

DISCUSSION AND CONCLUSION

- Complete-prefix reuse preserved success only when reuse became rare.
- Scene-camera reuse preserved the observed success count but did not improve latency relative to batched dense inference.
- Fine-grained K/V reuse reduced measured query time but failed all 120 cached closed-loop episodes.
- The tested strategies did not provide a reliable success and latency tradeoff.

LIMITATIONS

- One OpenVLA-OFT checkpoint
- One simulator benchmark family
- One GPU class
- Different populations across mechanisms
- Terminal task success only

REFERENCES

- [1] Kim et al. OpenVLA. CoRL 2024. [2] Kim et al. OpenVLA-OFT. RSS 2025.
[3] Liu et al. LIBERO. NeurIPS 2023. [4] Xu et al. VLA-Cache. NeurIPS 2025.

ACKNOWLEDGMENTS

Rutgers SRC Lab
Research guidance from Cheng Yang and Bo Yuan

CODE AND EVIDENCE

github.com/vwdr/SAVR

